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CIS Administration Application

The CIS Administration Application is the administration interface for Content Integration Suite (CIS). Deploying the CIS Administration Application separately from CIS allows administrators to choose to not install it on a production server for security purposes. The CIS Admin WAR file (cis-admin-10g.war) is included with the CIS distribution file. Follow your application server or servlet engine instructions to deploy the WAR file.

> **Note:** This guide uses the forward slash (/) to separate directories. Depending on your operating system, you may need to change the separation markers when defining directories.

This chapter contains the following sections:

- "Content Server Configuration" on page 2-1
- "Using the CIS Administration Application" on page 2-2
- "Enabling CIS Administration Application Security" on page 2-3
- "Command Caching" on page 2-5
- "Logging" on page 2-6

2.1 Content Server Configuration

You must configure the content server to communicate with CIS by enabling the IP address of the application server. You must also install and enable the CIS_Helper component on your content server instance.

> **Note:** Ensure that the CIS_Helper component on your content server instance is enabled. Refer to the Oracle Content Server administration documentation for more information.

Enabling the IP Address on the Content Server

You must enable the IP address of the application server. This allows the content server to listen for connections from the application server.

1. On your content server instance launch the System Properties editor:

   Select **Content Server** then *instance name* then **Utilities** and then **SystemProperties**.

2. Select the Server tab.

3. Enter your application server specific information for the **Hostname Filter** and **IP Address Filter**.

   The wildcard (*) is accepted, but IP addresses must take the form x.x.x.x regardless of wildcards, and must be separated by a vertical bar.

   Example: `12.34.56.*|12.34.57.*|12.35.*.*`

4. You must restart the content server for these changes to take effect.

2.2 Using the CIS Administration Application

After deploying the CIS Administration Application, you will need to perform several configuration steps to communicate with your content server instance and to validate the connection.

This section contains the following topics:

- "Modifying the Global Configuration" on page 2-2
- "Configuring an Adapter for Content Server" on page 2-2
- "Running CIS Validation Tests" on page 2-3

2.2.1 Modifying the Global Configuration

Once you have deployed the CIS Administration Application and restarted the application server, you can modify the global configuration.

1. Launch the CIS Administration Application.
2. Click **Adapter Configuration.**
3. Click **Modify Global Configuration**.
4. Enter the SCS Command Web URL (the web address clients use to communicate with the CIS servlets).

   Optional: Enter the Server JNDI Properties (the JNDI properties used by the server to make JNDI connections).

5. Click **Apply**.

If this is a new installation, you must also configure the adapters for content server to communicate with the application server.

2.2.2 Configuring an Adapter for Content Server

You must configure an adapter for each content server instance.

1. Launch the CIS Administration Application.
2. Click **Adapter Configuration**.
3. For new installations, click **Create New SCS Adapter**.
4. To update an existing installation; click the adapter name (e.g., myadapter) on the Available Adapters to Configure list.
5. On the Adapter Properties page, specify these properties:
   - **Name**: The name of the adapter (e.g., myadapter).

- **Adapter Type**: Type of communication transport. Select **Socket** for open communication directly through the content server or **Web** for communication via the web server which requires individual authentication for each request.
- **Host**: The name or IP address of the content server instance.
- **Port**: The content server port number (e.g., 4444).
- **Use Persistent Connections**: Enable this checkbox if you want to keep the connection alive after a request is completed.

  If you enable persistent connections, you also need to configure them on your content server instance (via the providers interface). Refer to the Oracle Content Server administration documentation for more information.
- **Connection Timeout**: When using persistent connections, this value is the maximum amount of time the connection will wait for response. Default is 20000 milliseconds.
- **Version**: The content server version. Choose **Default** if you are using version 7.5 or greater or 7.0 if you are using version 7.0.
- **Resource Pool Initial Size**: The initial size of the adapter connection resource pool. Default is 10.
- **Resource Pool Initial Size**: The maximum size of the adapter connection resource pool. Default is 20.

6. Click **Apply**.

After configuring the CIS adapter you may want to run a sample command such as a search. To do this, restart the application server and then see "Running CIS Validation Tests" on page 2-3 for more information.

2.2.3 Running CIS Validation Tests

Two validation tests for installed adapters are available by clicking the **Actions** icon:

- **Validate Mapped Directory Configuration**: This will only pass if the properties for Content Mapping Location have been validly set.
- **Validate Communication with Content Server**: Returns configuration validation results. If communication is validated the message, *Communication to the content server is functioning correctly* will display.

Figure 2–1 Actions Icon



Once you have validated your configuration, click the **View/Execute Installed Commands** and run some commands against the content server.

2.3 Enabling CIS Administration Application Security

By default, the CIS Administration Application has no security. Enabling security provides a login screen.

Note: The system administrator may need to modify these steps, depending on the configuration of the particular application server environment.

Follow these steps to enable security:

1. If you have not already done so, unbundle the CIS distribution file.
2. Unbundle the CIS Admin Web WAR file (e.g., cis-admin-web-10g.war depending on version number). To avoid confusion with other files, you may want to create a sub-directory named *unbundled-admin-web-war* and unbundle to that location.
3. In the WEB-INF directory, open the **web.xml** file in a text-only editor.
4. In the **web.xml** file, add security-constraint and security-role information. Depending on your application server, these entries may vary.

   Example:

```
<security-constraint>
  <web-resource-collection>
    <web-resource-name>staticfiles</web-resource-name>
    <url-pattern>/ui/resources/*</url-pattern>
  </web-resource-collection>
</security-constraint>

<security-constraint>
  <web-resource-collection>
    <web-resource-name>secure</web-resource-name>
    <url-pattern>/*</url-pattern>
  </web-resource-collection>
  <auth-constraint>
    <role-name>CISAdministrator</role-name>
  </auth-constraint>
</security-constraint>

<login-config>
  <auth-method>FORM</auth-method>
  <form-login-config>
    <form-login-page>/login.do</form-login-page>
    <form-error-page>/login.do</form-error-page>
</login-config>

<security-role>
  <description>CIS Administrator</description>
  <role-name>CISAdministrator</role-name>
</security-role>
```

5. Change the default security role-name entries (optional step).

```
<role-name>CISAdministrator</role-name>
```

   When you access the CIS Administration Application, you will encounter a login screen. Only those assigned the defined security role-name will be allowed access. You will need to create this role in your application server and assign it to the appropriate users. Consult your application server documentation for more details.

6. Save the **web.xml** file.
7. Rebundle CIS Admin Web WAR file (including the edited web.xml file). If you followed the steps above, this would be the contents of the *unbundled-admin-web-war* directory.

For WebSphere only, if you have enabled Global Security, do not check the Enforce Java 2 Security option.

2.4 Command Caching

CIS features a robust command caching mechanism that allows users to turn on/off caching for different commands and allows users to specify a timeout value for the cache and the size of the cache. The caching mechanism allows a command to be tied to a content server subject so that when a subject change happens, the command cache is expired. Finally, it provides the idea of either sharing the cache with all users or isolating the caching resources to a single user.

When a command is cached, the request is intercepted by the CIS layer and an already existing response for that request is returned through the system. This implies that there are less round-trips to the content server as many of the requests will be handled before reaching the content server.

All data is cached per command; there is no sharing of data between commands. The cache data can live in one of two caches: a user cache or a global cache. A user cache is a cache that only the current user can access. It is determined by the ICommand.getUser() return value; all command requests, for a given command, that return the same value will share the same cache.

This section contains the following topics:

- "Cache Expiration" on page 2-5
- "Enabling Command Caching" on page 2-5
- "Event Polling" on page 2-6

2.4.1 Cache Expiration

All cache values have a *time-to-live* in the cache. This value determines how long a cache is valid; the larger the value the better performing the cache but also the greater chance the cache will become full of items that are out of date. This value, specified in milliseconds, can be configured from the adapter configuration screen under the Command Cache Properties tab.

The value should be set high enough so the caching helps performance. Setting the value too low will cause CIS to spend too much time adding and removing items from the cache which will remove any performance benefit caching may provide. Alternatively, setting the value to high will create a greater frequency of having cache full of items that are out of date (a *dirty cache*).

2.4.2 Enabling Command Caching

Caching be enabled for any standard command from within the Active API. By default, caching is enabled on the file.getFileStream command (needed in conjunction with file caching) on the document.information.getInformationByID.

Follow these steps to enable command caching:

1. Launch the CIS Administration Application.
2. Click **Adapter Configuration**.
3. Click the adapter to configure (e.g., myadapter).
4. Select the **Command Cache Properties** tab.
5. Click the **View/Modify** actions link.
6. Click the **Create** actions link.
7. Select a command from the drop-down list.

8. Click **Select**.
9. Set these properties (or use the defaults):
   - **Cache Enabled**: Select **true** from the drop-down list to enable caching.
   - **Cache Type**: Select the type of cache: User (per user) or All (globally shared). The cache type should almost always be set to User as setting to All shares the same cache with all users.
   - **Cache Timeout**: Timeout of items in the cache in milliseconds (Default: 1800000).
   - **Cache Subjects**: Subjects from the content server on which this cache is dependent (documents, users, subscriptions, etc.).

     The cache subjects associate the Command Cache with subjects in the content server. By default, the subject is documents meaning that when any documents are updated in the content server (i.e., created, deleted, or modified) the cache for this command will be cleared.

     When it is determined the a content server subject has been updated, all cached commands corresponding to the subject will be flushed. Refer to the Oracle Content Server administration documentation for more information.
10. Click **Submit**.

2.4.3 Event Polling

The event polling mechanism in CIS is related to the Command Cache, in that it polls the content server on a scheduled interval and if changes are found, it sends out events through the CIS layer. These events tell the Command Cache that content has been updated, a subject has changed, etc., and to clear the cache appropriately. If the event poller is not enabled (via the Event Polling Properties tab on the adapter configuration page), then no events will be generated and the cached items will only be expired via their timeout values.

The EventPoller looks for two types of events:

- Subject events
- Document events (added/deleted/updated)

Subject events are indications by the content server that something has changed under a given subject. These subjects are items such as documents, users, subscriptions, etc. Any of these subjects can be used to dirty an item in the command cache. Refer to the Oracle Content Server administration documentation for more information

2.5 Logging

CIS uses the Apache Commons Logging framework for logging bundled with the log4j implementation of the Apache Commons framework. Thus, all logging calls are routed to the log4j system. The log4j system is controlled by an XML file that describes which log categories will be displayed.

Refer to the Apache documentation for more information:
`http://jakarta.apache.org/commons/logging/`

The *cis-logging.xml* log4j configuration file is located in the /conf directory of the unbundled distribution file. This directory also contains the adapterconfig.xml file and should be in your Classpath.

Using the defaults entries, only CIS warnings and some important information are displayed upon startup. All logging calls are routed to both the application System.out (i.e. to the console) and outputs to the /bin/scsadapter.log file:

Example (default entry):

```
<param name="File" value="scsadapter.log"/>
```

For logging in a servlet, add the following to the XML file:

```
<category name=my_servlet additivity=false>
  <priority value=debug/>
  <appender-ref ref=ALL/>
</category>
```

Logging Categories

This table provides some important categories for CIS logging. In general, the categories correspond to the class names where the log message originated.

Category	Description
com.stellent.cis	The root category; should catch all the CIS-related messages. Note that most messages are at a DEBUG or INFO level; set to WARN to only get actual warnings or errors.
com.stellent.cis.impl.CISApplication	Category used to print out the initialization banner. Set this to INFO to see the startup banner.
com.stellent.cis.server.api.scs	Category for all the content server interactions. Set this to INFO or DEBUG to see logging related to pooling, communication, etc.The DEBUG level returns the most logging.
com.stellent.cis.server.api.scs.protocol.impl.HDAProtocol.write	Category to see what is written to the content server.
com.stellent.cis.server.api.scs.protocol.impl.HDAProtocol.read	Category to see what is read from the content server.